ELEC 872 - Replication Lab 3: AffectNet

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1 Discussion

All experiments use FER+ labels (8 classes), and test-time augmentation on the 3376-image test set. I report macro-F1 as the main score and accuracy for comparison.

1.1 Baseline (ResNet-18, ImageNet)

I ran the default model, which fine-tunes an ImageNet-pretrained ResNet-18 on faces upscaled from 48×48 to 128×128. I got **accuracy 0.8483** and **macro-F1 0.7179**. Accuracy is much higher than macro-F1 because the common classes dominate, while rare ones are often wrong, which is why macro-F1 is the main metric.

1.2 48×48 Images Training

To train 48×48 resolution, adapted the network's first layer to small images: I replaced ResNet's 7×7 stride-2 convolution with a 3×3 stride-1 convolution and removed the first max-pool, so a tiny 48×48 image is not shrunk away before the residual blocks see it.

Model	Img. size	Accuracy	Macro-F1
ResNet-18 (baseline)	128	0.8483	0.7179
ResNet-18 small	48	**0.8563**	**0.7238**

Table 1: Performance comparison of ResNet-18 models with different image sizes.

The native 48×48 model is slightly better ($+0.6$ macro-F1) and cheaper to train, so upscaling was not helping accuracy here; it only added computation.

1.3 Improving Accuracy

To improve upon the baseline, I trained for 50 epochs with mixup augmentation ($\alpha = 0.2$) and an EMA with decay 0.999. Mixup linearly interpolates pairs of images and target label distributions, regularizing the network against overconfidence and smoothing decision boundaries across ambiguous facial expressions. EMA maintains a running average of the network weights across training steps, damping stochastic gradient oscillations and settling into flatter, more generalizable minima. The extended 50-epoch schedule gives the model more time to converge under this regularized training.

Model	Accuracy	Macro-F1
ResNet-18 baseline	0.8483	0.7179
+ EMA + mixup + 50 epochs	**0.8602**	**0.7292**

Table 2: Performance comparison of ResNet-18 with and without EMA and mixup.

This gained $+1.1$ macro-F1 and $+1.19$ accuracy over the baseline.

1.4 Pretraining

1.4.1 No Pretraining

I trained ResNet-18 from random initialization for 50 epochs, which loads the same architecture with no ImageNet weights. Even with 50 epochs I only got **accuracy 0.8430**, **macro-F1. 0.6985**; below the pretrained baseline, which shows the ImageNet features are genuinely useful.

1.4.2 AlexNet

I trained the AlexNet, with its final classifier layer swapped for an 8-class classification head for 30 epochs. I got **accuracy 0.8229**, **macro-F1 0.6873**, the lowest of the pretrained runs.

1.4.3 AlexNet vs. ResNet-18

I compared the training curves (training loss, validation loss, validation accuracy, and validation macro-F1) of both models are compared in Figure 1. From the beginning of training, ResNet-18 consistently outperformed AlexNet in terms of validation accuracy and macro-F1. Furthermore, ResNet-18 exhibited a much steeper decrease in both training and validation loss, indicating faster

convergence. In contrast, AlexNet showed a more gradual reduction in loss and consistently lagged behind ResNet-18 in both validation accuracy and macro-F1 throughout the validation.

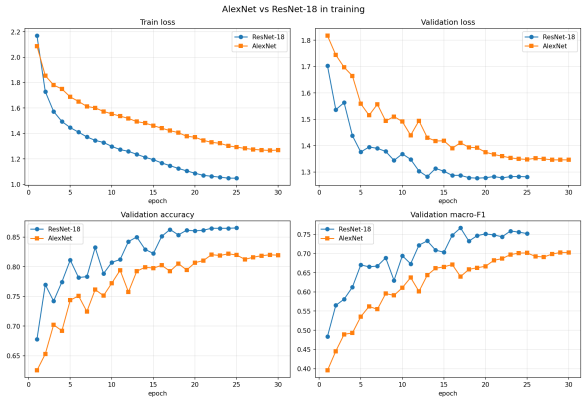


Figure 1: AlexNet vs. ResNet-18: train loss, val loss, val accuracy, val macro-F1.

1.4.4 Face-Pretrained Network (VGGFace2)

I trained InceptionResnetV1 pretrained on **VGGFace2**. VGGFace2 pretraining already encodes face-specific features, so they transfer far better to expression recognition than generic ImageNet features. It takes 160×160 inputs, so I trained at that resolution.

Pretraining	Model	Accuracy	Macro-F1
None (scratch)	ResNet-18	0.8430	0.6985
ImageNet	AlexNet	0.8229	0.6873
ImageNet	ResNet-18	0.8483	0.7179
Faces (VGGFace2)	Inception	**0.8655**	**0.7425**

Table 3: Performance comparison across different pretraining strategies and architectures.

VGGFace2 is the best overall (approx. $+2.5$ macro-F1 over the ImageNet baseline), so that means that domain-matched pretraining helps most.

1.5 Predicting My Own Image

The VGGFace2 model had the highest macro-F1, so I used it as my best model to predict the expression in my own photo. The model detects the face, crops it, and outputs the expression probabilities (Figure 2). So it seems correctly identified my facial expression.

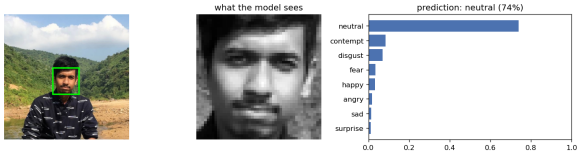


Figure 2: Example prediction from the VGGFace2 model.

2 Conclusion

Across all experiments, macro-F1 was the deciding metric, and the face-pretrained VGGFace2 network was the best model (macro-F1 0.7425, accuracy 0.8655), clearly ahead of the ImageNet ResNet-18 baseline (macro-F1 0.7179 and accuracy 0.8483). Three findings stand out. First, adapting the network to native 48×48 images matched and slightly beat upscaling to 128×128 at a lower computational cost, showing the upscaling added no real information. Second, mixup, EMA, and a longer 50-epoch schedule gave a reliable $+1.1$ macro-F1 gain over the baseline. Third, the pretraining study showed that the source domain matters most: face-domain pretraining beat generic ImageNet pretraining, which in turn beat both AlexNet and training from scratch. Throughout, the persistent gap between accuracy ($\sim 85\%$) and macro-F1 ($\sim 72\%$) reflects the class imbalance in FER2013, where rare expressions are the hardest to classify.

The code files are available in the AffectNet repository: https://github.com/yeasintaha/AffectNet.git